

Linear Transformation Questions

Friday, 26 April 2024 4:52 pm

① CONSIDER THE BASIS
 $S = \{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$ FOR $\mathbb{R}^3$,
 WHERE $\underline{v}_1 = (1, 1, 1)$, $\underline{v}_2 = (1, 1, 0)$,
 AND $\underline{v}_3 = (1, 0, 0)$.
 LET $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ BE THE
 LINEAR TRANSFORMATION
 SUCH THAT
 $T(\underline{v}_1) = (1, 0)$, $T(\underline{v}_2) = (2, -1)$,
 $T(\underline{v}_3) = (4, 3)$
 FIND A FORMULA FOR $T(x_1, x_2, x_3)$.
 THEN USE THIS FORMULA TO
 COMPUTE $T(2, -3, 5)$.

$$= \begin{bmatrix} x \\ 1 \\ 2 \end{bmatrix}$$

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 \vec{v}_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$c_1 + c_2 + c_3 = x \Rightarrow c_3 = x - (y - z) - z = \underline{x - y}$$

$$c_1 + c_2 = y \Rightarrow \underline{c_2 = y - z}$$

$$\underline{c_1 = z}$$

$$\Rightarrow T(c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 \vec{v}_3)$$

$$c_1 T(\vec{v}_1) + c_2 T(\vec{v}_2) + c_3 T(\vec{v}_3) \quad \therefore T \text{ is linear transform}$$

$$\Rightarrow c_1 \vec{w}_1 + c_2 \vec{w}_2 + c_3 \vec{w}_3$$

$$\Rightarrow c_1 \omega_1 + \dots$$

$$\Rightarrow z \begin{pmatrix} 1 \\ 0 \end{pmatrix} + (y-z) \begin{pmatrix} 2 \\ -1 \end{pmatrix} + (x-y) \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{bmatrix} z + 2(y-z) + 4(x-y) \\ 0 + (y-z)(-1) + 3(x-y) \end{bmatrix} = \begin{bmatrix} 4x - 2y - z \\ 3x - 4y + z \end{bmatrix}$$

$$\begin{bmatrix} 4 & -2 & -1 \\ 3 & -4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \dots \\ \dots \end{bmatrix}$$

$\uparrow$ $\mathbb{R}^3$

$$At (2, -3, 5)$$

$$\begin{bmatrix} 4 & -2 & -1 \\ 3 & -4 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 9 \\ 23 \end{bmatrix}$$

Q.no.2

12

CHECK WHETHER THE FOLLOWING MAPPINGS ARE LINEAR?

(a) $F(x, y) = (2x, y)$

(b) $F(x, y) = (x^2, y)$

(c) $F\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = b + c$

(d) $F\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x \\ x+y+z \end{bmatrix}$

(e) $T(A) = AB, A, B \rightarrow$ MATRICES

a) Let $u = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, v = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$

$\Rightarrow F(u+v) = F\left(\begin{bmatrix} x_1+x_2 \\ y_1+y_2 \end{bmatrix}\right) = (2(x_1+x_2), y_1+y_2)$

$$f(\underline{u} + \underline{v}) = \begin{matrix} \text{[} y_1 \text{]} & \text{[} y_2 \text{]} \\ f \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix} = \begin{bmatrix} 2(x_1 + x_2) \\ y_1 + y_2 \end{bmatrix} \\ = \begin{bmatrix} 2x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ y_2 \end{bmatrix} = F(\underline{u}) + F(\underline{v}) \end{matrix}$$

$$F(k\underline{u}) = \begin{bmatrix} 2(kx_1) \\ y_1 \end{bmatrix} = k \begin{bmatrix} 2x_1 \\ y_1 \end{bmatrix} = k F(\underline{u})$$

$$b) \quad f(\underline{u} + \underline{v}) = \begin{bmatrix} (x_1 + x_2)^2 \\ (y_1 + y_2) \end{bmatrix} = \begin{bmatrix} x_1^2 + 2x_1x_2 + x_2^2 \\ y_1 + y_2 \end{bmatrix}$$

$$F(\underline{u}) + F(\underline{v}) = \begin{bmatrix} x_1^2 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2^2 \\ y_2 \end{bmatrix} \quad \text{Not Equal because of } 2x_1x_2 \text{ expression}$$

Not Linear

$$c) \quad \text{Let } \vec{A} = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \text{ \& } \vec{B} = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$$

$$F(\vec{A} + \vec{B}) = F \begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{bmatrix} = b_1 + b_2 + c_1 + c_2$$

$$= (b_1 + c_1) + (b_2 + c_2) = F \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} + F \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = F(\vec{A}) + F(\vec{B})$$

$$F(k\vec{A}) = F \left(k \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \right) = F \begin{bmatrix} ka_1 & kb_1 \\ kc_1 & kd_1 \end{bmatrix}$$

$$= kb_1 + kc_1 \Rightarrow k(b_1 + c_1)$$

$$\Rightarrow k F \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \Rightarrow k F(\vec{A})$$

Hence it is linear

$$d) \quad F(\underline{u} + \underline{v}) = \begin{bmatrix} x_1 + x_2 \\ (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) \end{bmatrix}$$

$$= \begin{bmatrix} x_1 \\ x_1 + y_1 + z_1 \\ F(u) \end{bmatrix} + \begin{bmatrix} x_2 \\ x_2 + y_2 + z_2 \\ F(v) \end{bmatrix}$$

$$F(ku) = F\left(k \begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = F\left(\begin{bmatrix} kx \\ ky \\ kz \end{bmatrix}\right) = \begin{bmatrix} kx \\ kx + ky + kz \\ kx + ky + kz \end{bmatrix}$$

$$= k \begin{bmatrix} x \\ x + y + z \\ x + y + z \end{bmatrix} \Rightarrow kF(u)$$

It is linear

$$e) T(\vec{A} + \vec{C}) = (A + C)B = AB + CB = T(A) + T(C)$$

$$T(k\vec{A}) \Rightarrow (kA)B = k(AB) = kT(A)$$

Hence it is linear

Q.no.3

(a) IF $T(\underline{e}_1) = (1, 1)$,
 $T(\underline{e}_2) = (3, 0)$ AND
 $T(\underline{e}_3) = (4, -7)$ THEN
 FIND $T(1, 3, 8)$ PROVIDED
T IS LINEAR. ($T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$)

ANSWER: $T(1, 3, 8) = (42, -55)$

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(b) ALSO FIND $T(x, y, z)$
 BY USING INFORMATION IN **(3)**
PART (a).

$$T(c_1 e_1 + c_2 e_2 + c_3 e_3) \\
= c_1 T(e_1) + c_2 T(e_2) + c_3 T(e_3) \therefore T \text{ is linear}$$

c_1, c_2, c_3

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 3 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 4 \\ -7 \end{pmatrix}$$

$$\begin{bmatrix} c_1 + 3c_2 + 4c_3 \\ c_1 + 0 - 7c_3 \end{bmatrix} = \begin{bmatrix} x + 3y + 4z \\ x - 7z \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3 & 4 \\ 1 & 0 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{At } (1, 3, 8) \rightarrow \begin{bmatrix} 1 & 3 & 4 \\ 1 & 0 & -7 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix} = \begin{bmatrix} 42 \\ -55 \end{bmatrix}$$

ANS. $\begin{bmatrix} x + 3y + 4z \\ x - 7z \end{bmatrix}$

(C) USING PART (b) FIND THE
MATRIX OF LINEAR TRANSFORMATION.

ANS. $A = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 0 & -7 \end{bmatrix}$

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Q.no.4

Q.29, Q.30 P.375
(8TH ED.)

OR Q.30, Q.31 P.395 (7TH ED.)

Homework 6b: Linear Transformation

Question 27 (From Book)

Solution: (a) $(T_1 + T_2)(v + w) = T_1(v + w) + T_2(v + w)$

$$\begin{aligned}
 (T_1 + T_2)(v + w) &= T_1(v + w) + T_2(v + w) \\
 &= T_1(v) + T_1(w) + T_2(v) + T_2(w) \\
 &\therefore T_1 \& T_2 \text{ Linear Transformation} \\
 &= (T_1 + T_2)v + (T_1 + T_2)w \\
 (T_1 + T_2)\alpha v &= T_1(\alpha v) + T_2(\alpha v) \\
 &= \alpha T_1(v) + \alpha T_2(v) \\
 &= \alpha(T_1(v) + T_2(v)) \\
 &= \alpha(T_1 + T_2)(v)
 \end{aligned}$$

(b) $T_1(x, y) = (2y, 3x) \quad T_2(x, y) = (y, x)$

$$\begin{aligned}
 \text{(b) } T_1(x, y) &= (2y, 3x) \quad T_2(x, y) = (y, x) \\
 (T_1 + T_2)(x + y) &= T_1(x, y) + T_2(x, y) \\
 &= (2y, 3x) + (y, x) \\
 &= (3y, 4x)
 \end{aligned}$$

Question 28 (From Book)

(a) $F(x_1, y_1) = (a_1x_1 + b_1y_1, a_2x_1 + b_2y_1)$

Solution:

$$\begin{aligned}
 F(\alpha(x_1) + x_2, \alpha(y_1) + y_2) &= (a_1(\alpha(x_1) + x_2) + b_1(\alpha(y_1) + y_2), a_2(\alpha(x_1) + x_2) + b_2(\alpha(y_1) + y_2)) \\
 &= (a_1\alpha(x_1) + b_1\alpha(y_1), a_2\alpha(x_1) + b_2\alpha(y_1)) + (a_1x_2 + b_1y_2, a_2x_2 + b_2y_2) \\
 &= \alpha F(x_1, y_1) + F(x_2, y_2)
 \end{aligned}$$

b) $F(x_1 + x_2, y_1 + y_2) = (a_1(x_1 + x_2)^2 + b_1(y_1 + y_2)^2)$ It is nonlinear, one can verify by getting terms $2x_1x_2$ and $2y_1y_2$.

Question 29 (From Book)

Solution: Let any vector $v = \sum_{i=1}^n c_i v_i$

$$T(v) = T\left(\sum_i c_i v_i\right) = \sum c_i T(v_i) \quad \because T \text{ is Linear}$$

since each v_i on T is 0

$$T(v) = \sum c_i 0 = 0$$

Question 30 (From Book)

Solution: Let any vector $v = \sum_{i=1}^n c_i v_i$

$$T(v) = T\left(\sum_i c_i v_i\right) = \sum c_i T(v_i) \quad \because T \text{ is Linear}$$

since each v_i on T is v_i

$$T(v) = \sum c_i v_i = v$$

Q) 32 From Book

$$\begin{aligned} T(f + \alpha g) &= 5(f(x) + \alpha g(x)) + 3 \int_a^x (f(t) + \alpha g(t)) dt \\ &= 5f(x) + \alpha(5g(x)) + 3 \int_a^x f(t) dt + \alpha \left(3 \int_a^x g(t) dt \right) \\ &= 5f(x) + 3 \int_a^x f(t) dt + \alpha \left(5g(x) + 3 \int_a^x g(t) dt \right) \\ &= T(f) + \alpha T(g) \end{aligned}$$

[Q. NO. 5] (PREVIOUS CONCEPT)
REVISITED.
 IF $\underline{u}_1 = \underline{v}_1$ AND $\underline{u}_2$ ARE **EIGENVECTORS** OF **A** CORRESPONDING TO λ THEN
 PROVE THAT $\underline{v}_2 = \underline{u}_2 - \frac{(\underline{u}_2 \cdot \underline{v}_1) \underline{v}_1}{\|\underline{v}_1\|^2}$ IS
 ALSO AN **EIGENVECTOR** OF **A** CORRESPONDING TO λ .

Q5 From Lect. 25

Solution:

$$\begin{aligned} A(\underline{v}_2) &= \left(A\underline{u}_2 - \frac{(\underline{u}_2 \cdot \underline{v}_1)}{\|\underline{v}_1\|^2} \underline{v}_1 \right) = A(\underline{u}_2) - A\left(\frac{(\underline{v}_2 \cdot \underline{v}_1)}{\|\underline{v}_1\|^2} \underline{v}_1\right) \because A \text{ is Linear} \\ &= \lambda \underline{u}_2 - \frac{(\underline{u}_2 \cdot \underline{v}_1)}{\|\underline{v}_1\|^2} \lambda \underline{v}_1 = \lambda \left(\underline{u}_2 - \frac{(\underline{v}_2 \cdot \underline{v}_1)}{\|\underline{v}_1\|^2} \underline{v}_1 \right) \\ A(\underline{v}_2) &= \lambda \underline{v}_2 \end{aligned}$$

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[Q. NO. 6] **(14)**
 (a) IF $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ BE THE MATRIX TRANSFORMATION $T(\underline{x}) = A\underline{x}$, FIND **KER(T)** AND **RANGE OF T** ($R(T)$).
 (b) SEE THE DEFINITIONS OF **NULLITY(T)** AND **RANK(T)** (P. 378 8TH ED., P. 397 7TH ED.)

$A\underline{x} = \underline{0}$ Kernel
 Nullspace
 of a matrix
 A.
 $\underline{p} \dots T$

(P. 378 8TH ED., P. 397 7TH ED.)
 WHERE T IS A LINEAR TRANSFORMATION.

"A"
 Range T
 "Ax=b"

DEFINITION 11.7. The **kernel** of a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the set of all $\mathbf{x} \in \mathbb{R}^n$ such that $T\mathbf{x} = \mathbf{0} \in \mathbb{R}^m$.

$$\ker(T) = \{\mathbf{x} \in \mathbb{R}^n \mid T(\mathbf{x}) = \mathbf{0}\}$$

DEFINITION 11.9. The **image** or **range** of T is the set of all $\mathbf{y} \in \mathbb{R}^m$ such that $\mathbf{y} = T(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^n$.

$$\text{range}(T) = \{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{y} = T(\mathbf{x}) \text{ for some } \mathbf{x} \in \mathbb{R}^n\}$$

Now let A be the $m \times n$ matrix corresponding to a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$. Then

$$\begin{aligned} \ker(T) &= \{\mathbf{x} \in \mathbb{R}^n \mid T(\mathbf{x}) = \mathbf{0}\} \\ &= \{\mathbf{x} \in \mathbb{R}^n \mid A_T \mathbf{x} = \mathbf{0}\} = \text{Null space of } A_T \end{aligned}$$

$$\begin{aligned} \text{range}(T) &= \{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{y} = T(\mathbf{x}) \text{ for some } \mathbf{x} \in \mathbb{R}^n\} \\ &= \{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{y} = A_T \mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\} = \text{column space of } A_T \end{aligned}$$

Part(b)

DEFINITION

If $T: V \rightarrow W$ is a linear transformation, then the dimension of the range of T is called the **rank of T** and is denoted by $\text{rank}(T)$; the dimension of the kernel is called the **nullity of T** and is denoted by $\text{nullity}(T)$.

If A is an $m \times n$ matrix and $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is multiplication by A , then we know from Example 1 that the kernel of T_A is the nullspace of A and the range of T_A is the column space of A . Thus we have the following relationship between the rank and nullity of a matrix and the rank and nullity of the corresponding matrix transformation.

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(c) USING PARTS (a & b) TRY THE FOLLOWING

IF A IS AN $m \times n$ MATRIX AND $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ IS MULTIPLICATION BY A , THEN

(i) NULLITY(A) = NULLITY(T)

(ii) RANK(A) = RANK(T)

(d) USING PART (a), TRY THE FOLLOWING:

LET $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ BE THE LINEAR OPERATOR DEFINED BY THE FORMULA

$$T(x_1, x_2, x_3)$$

$$= (x_1 + x_2 + 2x_3, x_1 + x_3, 2x_1 + x_2 + 3x_3)$$

FIND BASES FOR THE KERNEL AND RANGE OF T .

HINT: WRITE DOWN

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_2 + 2x_3 \\ x_1 + x_3 \\ 2x_1 + x_2 + 3x_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = AX$$

AND PROCEED.

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Part(c)

If A is the matrix representation of a linear transformation T , then

1. $\mathcal{N}(T) = \mathcal{N}(A)$ and $\mathcal{R}(T) = \mathcal{R}(A)$.
2. The nullity of T is the same as the nullity of A .
3. The rank of T is the same as the rank of A .

Part(d)

for finding out the kernel we have to do

$$T(x_1, x_2, x_3) = (0, 0, 0)$$

$$\text{so } x_1 + 2x_3 + x_2 = 0, x_1 + x_3 = 0, 2x_1 + x_2 + 3x_3 = 0$$

$$\text{so } x_1 = -x_3$$

implies also we get, $x_2 + x_3 = 0$

$$\text{so } (x_1, x_2, x_3) = (-x_3, -x_3, x_3)$$

so dimension of kernel is 1 and bases is $\{(-1, -1, 1)\}$

and for range we know we have to collect linearly independent columns

$$\text{here columns are } \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$\text{so } \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

so linearly independent columns is first and second column

$$\text{so basis for range is } \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

dimension of range space is 2

Q. no. 8

SEE THE STATEMENT OF THE
DIMENSION THEOREM FOR
LINEAR TRANSFORMATIONS
ON $\left\{ \text{P. 379 (8TH ED.) OR} \right.$
 $\left. \text{P. 398 (7TH ED.)} \right\}$

THEN

PROVE THAT THIS THEOREM
AGREES WITH DIMENSION
THEOREM FOR MATRICES

IF $[A]$ IS AN $m \times n$ MATRIX
AND $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ IS MULTIPLI-
CATION BY [A].

Question 08:

Let A denote an $m \times n$ matrix of rank r and let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ denote the corresponding matrix transformation given by $T_A(\mathbf{x}) = A\mathbf{x}$ for all columns $\mathbf{x}$ in $\mathbb{R}^n$. It follows from Example 7.2.1 and Example 7.2.2 that $\text{im } T_A = \text{col } A$, so $\dim(\text{im } T_A) = \dim(\text{col } A) = r$. On the other hand Theorem 5.4.2 shows that $\dim(\ker T_A) = \dim(\text{null } A) = n - r$. Combining these we see that

$$\dim(\text{im } T_A) + \dim(\ker T_A) = n \quad \text{for every } m \times n \text{ matrix } A$$

The main result of this section is a deep generalization of this observation.

Theorem 7.2.4: Dimension Theorem

Let $T : V \rightarrow W$ be any linear transformation and assume that $\ker T$ and $\text{im } T$ are both finite dimensional. Then V is also finite dimensional and

$$\dim V = \dim(\ker T) + \dim(\text{im } T)$$

In other words, $\dim V = \text{nullity}(T) + \text{rank}(T)$.

Proof. Every vector in $\text{im } T = T(V)$ has the form $T(\mathbf{v})$ for some $\mathbf{v}$ in V . Hence let $\{T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_r)\}$ be a basis of $\text{im } T$, where the $\mathbf{e}_i$ lie in V . Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\}$ be any basis of $\ker T$. Then $\dim(\text{im } T) = r$ and $\dim(\ker T) = k$, so it suffices to show that $B = \{\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{f}_1, \dots, \mathbf{f}_k\}$ is a basis of V .

1. B spans V . If $\mathbf{v}$ lies in V , then $T(\mathbf{v})$ lies in $\text{im } T$, so

$$T(\mathbf{v}) = t_1 T(\mathbf{e}_1) + t_2 T(\mathbf{e}_2) + \dots + t_r T(\mathbf{e}_r) \quad t_i \in \mathbb{R}$$

This implies that $\mathbf{v} - t_1 \mathbf{e}_1 - t_2 \mathbf{e}_2 - \dots - t_r \mathbf{e}_r$ lies in $\ker T$ and so is a linear combination of $\mathbf{f}_1, \dots, \mathbf{f}_k$. Hence $\mathbf{v}$ is a linear combination of the vectors in B .

2. B is linearly independent. Suppose that t_i and s_j in $\mathbb{R}$ satisfy

$$t_1 \mathbf{e}_1 + \dots + t_r \mathbf{e}_r + s_1 \mathbf{f}_1 + \dots + s_k \mathbf{f}_k = \mathbf{0} \quad (7.1)$$

Applying T gives $t_1 T(\mathbf{e}_1) + \dots + t_r T(\mathbf{e}_r) = \mathbf{0}$ (because $T(\mathbf{f}_i) = \mathbf{0}$ for each i). Hence the independence of $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_r)\}$ yields $t_1 = \dots = t_r = 0$. But then (7.1) becomes

$$s_1 \mathbf{f}_1 + \dots + s_k \mathbf{f}_k = \mathbf{0}$$

so $s_1 = \dots = s_k = 0$ by the independence of $\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$. This proves that B is linearly independent. □

Note that the vector space V is not assumed to be finite dimensional in Theorem 7.2.4. In fact, verifying that $\ker T$ and $\text{im } T$ are both finite dimensional is often an important way to *prove* that V is finite dimensional.

Note further that $r + k = n$ in the proof so, after relabelling, we end up with a basis

$$B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_r, \mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$$

of V with the property that $\{\mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ is a basis of $\ker T$ and $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_r)\}$ is a basis of $\text{im } T$. In fact, if V is known in advance to be finite dimensional, then *any* basis $\{\mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ of $\ker T$ can be extended to a basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_r, \mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ of V by Theorem 6.4.1. Moreover, it turns out that, no matter how this is done, the vectors $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_r)\}$ will be a basis of $\text{im } T$. This result is useful, and we record it for reference. The proof is much like that of Theorem 7.2.4 and is left as Exercise 7.2.26.